AICT

Assignment # 1

OSI and TCP/IP

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OSI Model:-

The OSI (Open System Interconnection) Model, developed by the International Organization for standardization, is a seven-layer conceptual frame work that standardizes how data is transmitted across a network. It has 7-Layers.

1-Physical Layer:-

It transmits raw binary data (bits) over physical media such as cables and fiber optics. It defines hardware elements like voltage levels, connectors, and transmission speeds.

2- Data Link Layer:-

It ensure reliable node-to-node data transfer using frames and MAC addresses. It detects and may correct error that occur in the Physical Layer.

3- Network Layer:-

It manages logical addressing (IP addresses) and determines the best Path for data routing. It ensures Packets travel from the source network to the destination network.

4- **Transport Layer:-**

It provides end-to-end communication and ensures reliable data delivery. It perform error detection, flow control, and segmentation of data.

5- **Session Layer:-**

It establishes, maintains, and terminates and communication session between devices. It manages synchronization and session checkpoints during data exchange.

6- **Presentation Layer:-**

It translates data formats between systems to ensure compatibility. It also handle encryption, decryption and data compression.

7- **Application Layer:-**

It provides network services directly to end-user applications. It enables services like web browsing, email and file transfers.

TCP/IP:-

The TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a practical networking framework developed by the Defense Advanced Research Projects Agency (DARPA) for communication over interconnected networks, including the internet.

It is the foundation of modern Internet communication.

It has 4 Layers.

1- Application Layer:-

Provides network services to users application such as web browsing and email clients. It combines the functions of Application, Presentation, and Session Layer of the OSI model.

2. Transport Layer:-

Ensures end-to-end communication between devices.
Use Protocols like (TCP) reliable and UDP (fast but unreliable).

3. Internet Layer:-

Handles Logical addressing and routing using IP models addressing and routing using IP addresses. Determines the best path for data transmission across networks.

4- **Network Access Layer:-**

Manges physical transmission of data over hardware. Includes MAC addressing and Interaction with network devices.